

Earth's Coordinates

Guided Notes

Driving question: How do we locate things on Earth?

1. Why we need a coordinate system

Directions given by landmark (“two miles past the old barn”) only work if you already know the landmark. A **coordinate system** gives every point on Earth one unique address built from a grid of _____ lines.

The grid is anchored to something that never moves: Earth's _____. The poles and the equator come from the way Earth spins, not from a human decision.

2. Latitude and longitude

Complete the table.

	Latitude	Longitude
Lines run...	_____	_____
Measured...	_____	_____
Do they touch?	_____	_____
Measured from...	_____	_____
That line is...	0° latitude	0° longitude
Range	0° to _____ N or S	0° to _____ E or W

Every coordinate needs a **number AND a direction**. 43° is not a location. 43° N is a line; 43° N, 76° W is a point.

Latitude is always written first.

Why does latitude stop at 90° but longitude go to 180°?

3. Degrees and minutes

When we zoom in, latitude and longitude are given in degrees *and* **minutes**, written with a prime symbol (').

$$1^{\circ} = \underline{\hspace{2cm}}' \quad 30' = \frac{1}{2}^{\circ} \quad 10' = \frac{1}{6}^{\circ}$$

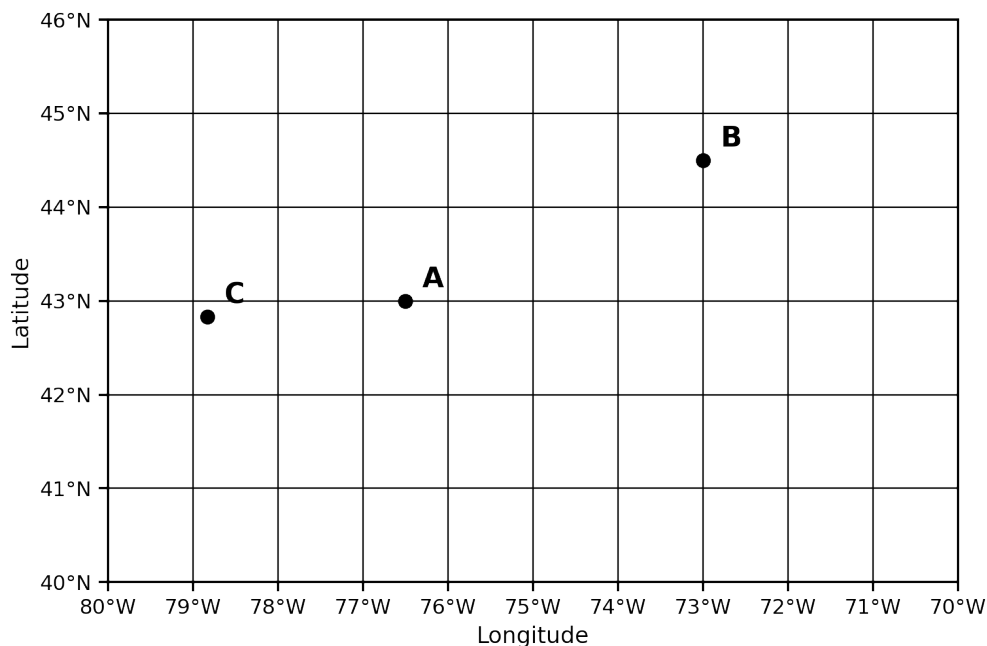
- Halfway between two degree lines is about _____'.
- We estimate minutes to the nearest _____ minutes.

Watch out: 76.5° is **not** $76^{\circ}50'$. Half a degree is _____', just like half an hour is 30 minutes.

Worked example. Syracuse, NY sits on the 43° parallel and a little past the 76° meridian: _____

4. Practice: reading the grid

Give the latitude and longitude of each point to the nearest 10 minutes. Remember the hemisphere letters.



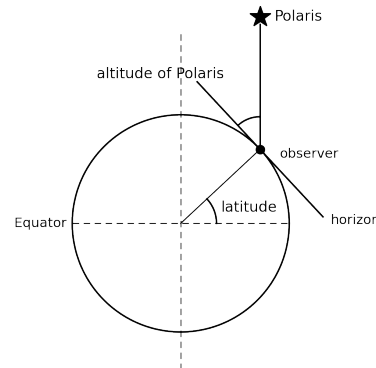
	Latitude	Longitude
A	_____	_____
B	_____	_____
C	_____	_____

5. Latitude and Polaris

Polaris — the **North Star** — sits almost directly above Earth's _____.

Because of that, the **altitude** of Polaris (its angle above the horizon) is equal to the observer's _____.

Polaris can only be seen from the _____ Hemisphere.



Check the extremes. Fill in the altitude of Polaris at each location.

Where you are	Latitude	Altitude of Polaris
North Pole	90° N	_____
Albany, NY	42° N	_____
Equator	0°	_____
Southern Hemisphere	—	_____

Moving around. Circle or fill in what happens to the altitude of Polaris.

- Travel **north**: Polaris gets _____ and latitude _____.
- Travel **south**: Polaris gets _____ and latitude _____.
- Travel **east or west**: Polaris _____ and latitude _____.

Moving 1° of latitude changes the altitude of Polaris by exactly 1° — about 111 km (69 miles) of travel on the ground.

6. Longitude and time zones

Meridians are counted east and west of the prime meridian until they meet at _____°, which is called the _____.

$$\frac{360^\circ \text{ of rotation}}{24 \text{ hours}} = \text{_____}^\circ \text{ per hour}$$

Time zones are _____° of longitude wide, and each one represents _____ of time.

Earth rotates from west to east, so places to the **east** rotate into sunlight first and their clocks are _____. Places to the **west** have clocks that are _____.

Worked example. Albany is 74° W. Denver is 105° W. How many hours apart are their clocks, and which is earlier?

$$105^\circ - 74^\circ = \underline{\hspace{2cm}} \quad \frac{31^\circ}{15^\circ/\text{h}} \approx \underline{\hspace{2cm}}$$

Denver is *west* of Albany, so Denver's clock is _____.

7. Check for understanding

1. The angle from the horizon to Polaris is equal to the _____ of the observer. [1]
 - A. latitude
 - B. longitude
2. Meridians are 15° apart, so time zones are separated by [1]
 - A. 4 hours
 - B. 24 hours
 - C. 1 hour
3. A ship sails due east across the Atlantic Ocean. The altitude of Polaris will [1]
 - A. increase
 - B. decrease
 - C. stay the same
4. How many minutes are in 1° of latitude? [1]
 - A. 10
 - B. 60
 - C. 100

Exit ticket

1. Using the *Generalized Bedrock Geology of New York State* map in your ESRT, give the latitude and longitude of Albany, NY to the nearest 10 minutes. [2]
2. An observer measures the altitude of Polaris as 30° . What is their latitude, and which hemisphere are they in? Explain how you know the hemisphere. [2]

3. Explain in one sentence why time zones are 15° of longitude wide.

[2]

Vocabulary

Latitude	angular distance north or south of the equator, 0° to 90°
Longitude	angular distance east or west of the prime meridian, 0° to 180°
Equator	the 0° line of latitude, halfway between the poles
Prime meridian	the 0° line of longitude, through Greenwich, England
Parallel	another name for a line of latitude
Meridian	another name for a line of longitude
Altitude	the angle of an object above the horizon
Int'l Date Line	the 180° meridian, where the calendar date changes